1 Introduction

Main points:

- Define SPE's
- Define Coexistence
- Growing interest in the literature
- tie to climate change
- basic sentence about how it works? (ie weaker competitiors go earlier)
- BUT. it doesn't work because **phenology** is not a trait
- in this paper we....say what we will do. One goal Goal demonstrate a simple way to better incorporate realistic physiology into models of coexistence and to integrate experiments and models

2 How do phenological differences arise?

- Discuss cues
- define phenological sensitive
- state that interaction between sensitivity and environment result in **realize phenologyical difference**
- Explain figure 1 and use as example

3 How we (currently) model phenology is coexistence (and why it doesn't work)

- a paragraph discussing how it's been done
- a paragraph about non-stationarity and shifting environmental means and why the current approaches break down under the conditions

4 A new hope

not sure if this should be its own section or transition from the previous one

- we need a model to do this
- here we show one of the simplest ways to do it by adding an intermediate between chillign nad response
- maybe a brief description of the apporach with refrence to supporting information for details
- even with such a simple approach we see really interesting things emerge about both the relationships between env., phenological sensitivities, and realized phenology as well as between phenological sensitivity and coexistence

Snappy title about results and inference

4.1 A shifting environment shifts the relationship between phenogical sensitivity difference and realized phenology differences

- Basically discuss a 3 panel figure. (histograms of chilling distrubutions and current figures 2c and d)

4.2 A shifting environment shifts the tradeoff between phenological sensitivity and competition

- The trade off between realize phenology and Rstar is the same
- Shift in the tradeoff between phenosensitivity and Rstar
- explicitly highlight that this is means different pairs of species coexist

5 A path forward

Our model is only the beginning, and while it is a noteworthy advance worthy of being published in a nature subsidiary or equivalently high impact journal, there is still a long way to go to forcast these dynamics. Here are some critical aspects. **I think I need help here. All of my thoughts are pretty germination/chilling specific. I could use some idea for contextualizing this more broadly in phenology, and coexistence**

1. multiple interacting cues
2. add risk of being early
3. trait and density dependent mechanisms (rudoff paper)/ ie, we vary germination fraction and phenology.
4. other life histories, i.e, perennials, trees.

Figures

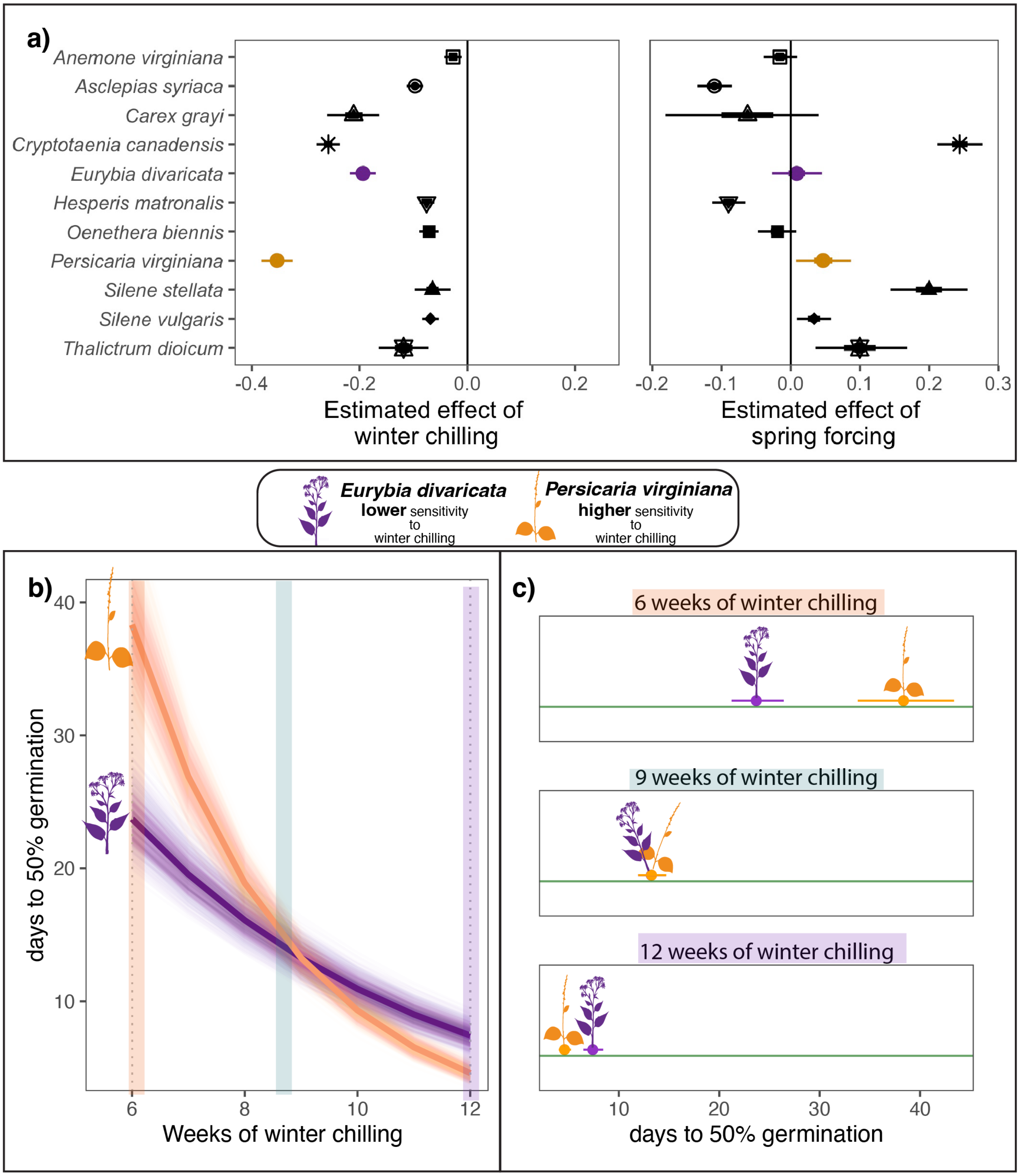

Figure 1: **Co-occur species can differ substantially in their phenological sensitivity (Δ phenological event/Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to winter chilling and spring forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 95% uncertainty intervals. In b) and c) we highlight the phenological sensitivity to winter chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) Depicts the sensitivity of each species to winter chilling, with thick lines representing the mean estimate and lighter lines representing modeled uncertainty. Panel c) illustrations how the species' different sensitivities manifest in contrasting realized germination differences under different climate conditions. Point represent the predicted mean number of days to 50% germination and bars the 95% uncertainty intervals

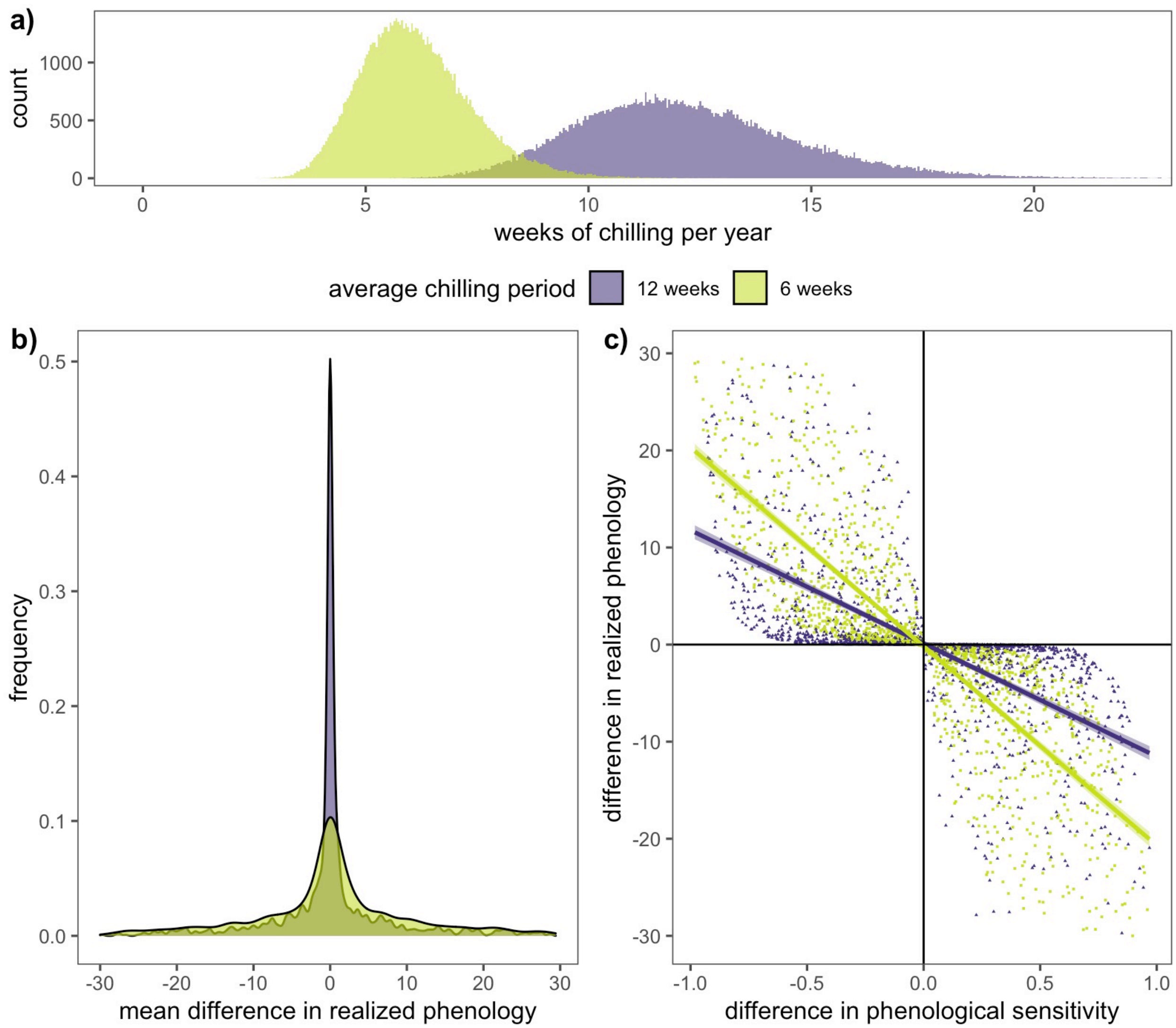

Figure 2: **Climate change will the relationship phenological sensitivity differences and realized phenological differences.** Panel a) shows how climate change will shift the distribution of the duration of chilling ecological communities receive each winter. Panel b) shows the frequency distributions for the average difference in realized germination phenology under both climate scenarios—shorter chilling durations result in larger realized phenological differences between species. These changes occur because the relationship between phenological sensitivity differences and realized phenology differences, depicted in panel c), are dependent on the environment, and a change in chilling conditions alters this relationship.

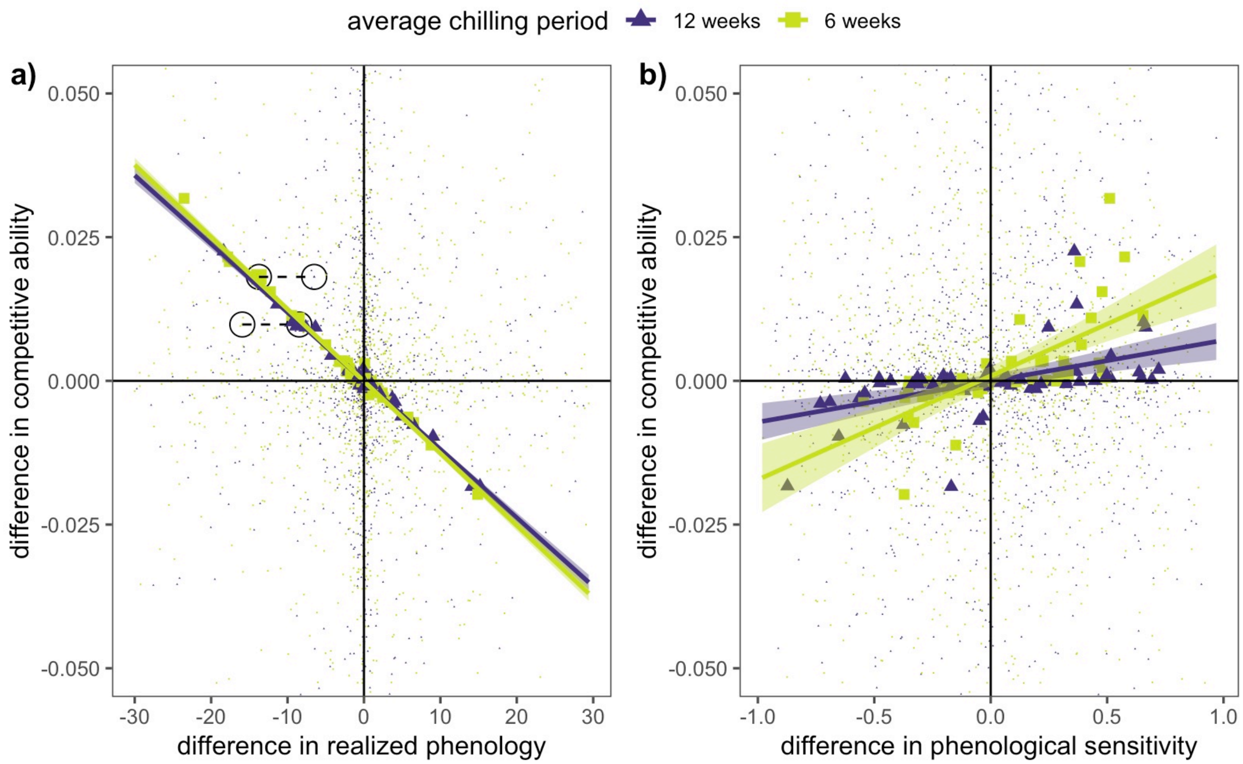

Figure 3: **Climate change will alter the relationship competititive ability, phenological sensitivty, realized phenology and coexistence.** Panel a) depicts the trade-off between realize phenological differences and competitive differences. While the tradeoff relationship between realized phenology and competition does not shift with climate change, species that coexisted under one climate scenario will not coexist if climate conditions shift, as highlighted by the two sets species pairs circled in panel a). Panel b) depicts the trade-off between phenological sensitivity differences and competitive differences under two climate scenarios (average of 12 and 6 weeks of chilling). With climate change, the trade-off between phenological sensitivity and competition does shift because less chilling amplifies the realized phenological differences among species